

Cross-Domain and Multilingual Fake Review Detection Using Machine Learning Models

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Abstract—User-generated content has a critical role in today’s digital marketplace, with online reviews influencing consumer choices and brand ratings. This influence has created deceptive content, designed to manipulate opinions and damage competitor reputations. Unfortunately, existing fake review detection models are mostly in one language and domain-specific. They perform well within their own domain or language but struggle with keeping a good performance when applied on new unseen data having different categories or languages. Moreover, their lack of transparency leaves users and platforms unable to verify the reasoning behind a certain prediction. This paper introduces CME, a Cross-domain, Multilingual and Explainable model, that detect fake reviews online. The model uses a multilingual transformer, specifically XML-RoBERTa, because of its training across multiple languages and domains. The model is designed to explore how multilingual and cross-domain representations perform on unseen data, allowing a more realistic evaluation of robustness across diverse inputs. Compared to existing models, CME includes an attention-based explainability layer into the prediction process, which makes visual explanations by highlighting the terms in a review that had the biggest impact on the decision. CME is trained on multilingual and multi-domain data to broaden the patterns it can learn and to evaluate how these representations behave when faced with a completely new domains. Its performance provides insights into the real challenges of cross-domain and multilingual fake review detection.

Index Terms—fake reviews, multilingual, cross domain, XML-RoBERTa, attention, explainability

I. INTRODUCTION

Online reviews have become an important part in how consumers make decisions, evaluate products, services, and businesses on online platforms such as e-commerce, hospitality, and many more. However, it has become very easy today to manipulate customers and add reviews on any website. In return, these fabricated opinions affect consumer trust, harm business reputation, and lower the reliability of recommendation systems. This problem has been worsening these days due to AI tools such as Chat-GPT, which easily generate a lot of realistic fake reviews. Earlier detection models were based on linguistic or behavioral features analyzed with classical machine-learning classifiers such as Support Vector Machines and Logistic Regression [1]. With the rise of deep learning architectures such as the CNN-LSTM model proposed by Bathla et al. [2] and transformer-based detectors like DeceptiveBERT [3], they gained more accuracy due to capturing contextual semantics. However, these also remain confined to English and domain-specific datasets, for example Yelp or Amazon, and are hardly interpretable. Subsequent research introduced after explainability using SHAP values, but such frameworks

are computationally expensive and fail to generalize across languages or domains. Similarly, while COAT [4] improved cross-domain adaptability and AraBERTv2 [5] extended the detection to Arabic, neither of them offered transparency nor multilingual scalability. To fill these gaps, this research proposes CME, a Cross-domain, Multilingual, and Explainable framework for fake-review detection. The CME uses the multilingual transformer XML-RoBERTa to capture semantic representations across more than 100 languages, and integrates cross-domain adaptability for robustness, and introduces an attention-based explainability layer that specifies the most influential words during prediction.

Unlike existing work that focuses separately on improving domain generalization, multilingual performance, or explainability using SHAP, none have integrated these aspects into a single approach, while using attention-based explainability instead of an additional framework. The CME framework, however, is designed to unify these capabilities. This eliminates the need for computationally heavy tools like SHAP to justify why the model has produced a certain output for a specific review.

The contributions of the work can be summarized as follows:

- CME offers a unified framework for multilingual fake review detection. This feature ensures the model’s precision during switching between languages, therefore letting the system catch fake reviews reliably across many languages, such as English and Arabic.
- The model’s architecture includes cross-domain training, allowing evaluation of how transformer-based models behave when exposed to different domains such as hotel, e-commerce, clothing, and travel-related reviews. This addition enables a realistic assessment of domain shift rather than assuming stable performance across categories.
- The integration of an attention-based explainability layer provides clear visual evidence for each prediction, helping users understand why a review is labeled as fake. Unlike slow methods like SHAP, our model produces attention scores as it reads the review, allowing it to instantly highlight the specific words and phrases that influenced the decision, which makes the system more transparent and trustworthy.
- A new evaluation dataset is assembled to test the model in realistic, zero-shot conditions. This dataset brings

together reviews from different sources, including a domain the model never saw during training. It provides a more practical way to assess how the model behaves when faced with unfamiliar inputs and helps uncover where cross-domain and cross-lingual generalization breaks down. This unified dataset will allow for consistent evaluation of CME’s adaptability and accuracy across different domains.

The rest of the paper is organized as follows. The literature review is presented in Section II. The system model is provided in Section III. The proposed approach is detailed in Section IV. The Results and Analysis of the model is elaborated in Section V. The Conclusion is formulated in Section VI

II. LITERATURE REVIEW

Many researches have studied fake review detection using methods that range from basic machine learning to deep learning and transformer-based models. Kwon et al. [5] showed that detecting fake text is becoming harder as AI-generated and misleading content sound more natural, while Liu et al. [6] reviewed existing methods and noted issues such as biased datasets, not a lot of language coverage, and weak generalization to new domains. A recent survey [7] and other overview works [8], [9] emphasized that most systems are still evaluated on small benchmarks and often ignore cross-domain and multiple languages. Several studies focused on linguistic and behavioral features. Models such as CNN-LSTM for sentiment and text patterns, such as Bathla et al. [4] and Abd et al. [10], and behavior-aware methods using rating averages and posting patterns [11] showed that combining what users write with how they behave improves accuracy. Ensemble and weighted stacking approaches [12] were also proposed to handle imbalanced datasets more robustly. More recent work fused features at a deeper level: WF-CFRB combines BERT-based contextual representations with reviewer behavioral features through a weighted fusion mechanism to outperform text-only models on Yelp data, while PosAtt-BiLSTM improves sequence modeling using positional encodings and a hybrid attention mechanism for long reviews. Transformer-based models have become central in this area. DeceptiveBERT [13] and BERT-ELECTRA combinations [14] improved semantic understanding of reviews, and works by Shunxiang et al. [15], Sedighi et al. [16], Maurya et al. [17], and Fahfouh et al. [1] added attention, sentiment-intensity, or dimensionality reduction (PCA/LDA) to better capture contextual patterns. Salminen et al. [18] integrated psycholinguistic features such as emotional exaggeration and cognitive load with transformer models, showing that these cues help separate fake from real reviews more effectively. Cross-language detection has also been explored: an ERNIE- and SGAN-based model [19] targets fake reviews across languages, aiming to improve robustness beyond standard monolingual BERT setups. Multilingual and cross-domain detection remain limited. Alhayan et al. [20] used AraBERTv2 for Arabic reviews, while Al-Zoubi et al. [21] combined multilingual BERT embeddings with a Weighted SVM for English, Spanish, and Arabic

data. Node-embedding methods model reviews as graphs: a Node2Vec-based approach achieved high accuracy on OpSpam and YelpChi and used SHAP and LIME for explainable predictions. Multimodal work has also appeared: Hou et al. [22] proposed FRIDRC, which uses both text and visual context (video frames or images) to detect fake reviews and reports very high F1 scores. Although some models ([2], [14], [23]) use SHAP or other tools for interpretability, these are often computationally expensive and still mostly restricted to single domains. Overall, existing research shows strong progress in linguistic, behavioral, transformer-based, graph-based, and multimodal approaches, but important gaps remain in multilingual coverage, cross-domain generalization, dataset diversity, and practical explainability—motivating models that are cross-domain, multilingual, and more transparent.

As shown in Table I, most studies did a good job with analyzing linguistic features, but they were limited to just one domain and one language. Only some, like COAT [4] and the work by Sedighi et al. [21], adapted their models across different domains. Models that handled multiple languages, such as AraBERTv2 [5] and [11], were more focused on specific languages, specifically Arabic only and Italian only, rather than multiple languages at once. When it comes to explainability, only Abedin et al. [10] and [11] applied SHAP for their analysis. None of the existing approaches used attention-based interpretability. More recent works expanded the landscape slightly, such as WF-CFRB [8] and FRIDRC [26], which unified multiple feature sources, and a Node2Vec-based method [9] that introduced SHAP explainability, but these still lacked multilingual or cross-domain strength. ERNIE-SGAN [7] addressed cross-language detection but did not provide interpretability or domain coverage. The proposed CME model aims to fill these gaps by combining cross-domain adaptability, multilingual support, and attention-based explainability into a single unified model.

III. SYSTEM MODEL

As shown in Figure 1, we consider the CME model which takes multilingual text and with some metadata as input and produces a classification label, fake or real, with an explanation as output.

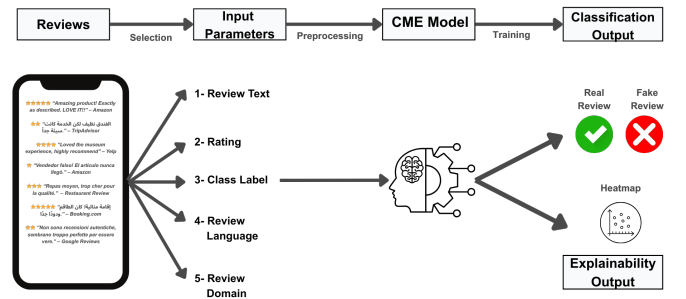


Fig. 1. High-level system model of CME showing input and output flow.

TABLE I
QUALITATIVE COMPARISON OF PREVIOUS STUDIES AND THE PROPOSED CME FRAMEWORK.

Studies	Linguistic modeling	Cross-domain	Multilingual	SHAP explainability	Attention-based explainability	Unified model
[13]	Yes	Yes	No	No	No	No
[2]	Yes	No	No	No	No	No
[3]	Yes	No	No	No	No	No
[4]	Yes	Yes	No	No	No	No
[5]	Yes	No	No	No	No	No
[6]	Yes	No	No	No	No	No
[10]	Yes	No	No	Yes	No	No
[11]	Yes	No	No	Yes	No	No
[14]	Yes	No	No	No	No	No
[21]	Yes	Yes	No	No	No	No
[24]	Yes	No	Yes	No	No	No
[25]	Yes	No	No	No	No	No
[19]	Yes	Yes	No	No	No	No
[8]	Yes	No	No	No	No	Yes
[26]	Yes	No	No	No	No	Yes
[9]	No	Yes	No	Yes	No	No
Proposed method	Yes	Yes	Yes	Yes	Yes	Yes

A. Input

The input to the system contains review texts of many languages (English and Arabic) and a mix of domains.

The first dataset, *Balanced-Data.xlsx*, includes all Arabic reviews: 300 hotel reviews, 700 product reviews and 700 restaurant reviews. The features of this dataset are reviewer demographics (Age, Gender), the name of the hotel/platform/restaurant the user is reviewing, the review text, a rating, polarity (if the review is a positive or a negative one), and class which labels the review as fake or real.

The second dataset, *deceptive-opinion.csv*, is the well-known Deceptive Opinion Spam Corpus, containing 1,600 English hotel reviews from TripAdvisor having as features the review text, the hotel name, polarity, and the truth label (truthful or deceptive).

The third dataset is *fake-review-dataset.csv* and it contains over 40,000 English Amazon product reviews across multiple categories, labeled as genuine or deceptive, providing a diverse e-commerce domain for cross-domain testing. The parameters of this dataset are rating, the text of the review, and the label of the review CG meaning computer generated and OR meaning original.

Datasets are not organized in the same way. Therefore, the most important parameters will be extracted from each, making them consistent. The common parameters are the following:

- 1) Review text
- 2) Rating (if possible, -1 if not)
- 3) Class label (False:1 or Real:0)
- 4) Language of the review
- 5) Domain of the review

In addition to the training datasets, a testing dataset was constructed to test the model under zero-shot conditions, which means tested on a domain that has never been seen in training. This dataset includes English reviews from clothing platforms, containing both fake and real reviews.

B. Output

The output of the system is twofold:

- 1) A binary classification that labels each review as Fake or Real.
- 2) A bar plot explanation of why the review was classified the way it was, meaning which words/terms contributed most to this classification. This allows users to understand why certain reviews are labeled fake or real based on the decision of the model.

IV. PROPOSED APPROACH

In this section, we propose multiple stages, each one designed to make sure that the model can operate effectively across domains, handle multiple languages, and provide transparent explainability.

A. Data Collection

The first stage consists in getting multilingual datasets and preparing them by preprocessing. Reviews will be collected from publicly available sources, including the Deceptive Opinion Spam Corpus (English), Arabic Hotel/Product/Restaurants Reviews Dataset, and an English dataset containing Amazon product reviews. The datasets contain labeled samples indicating whether a review is originally fake or not, along with other parameters that will be used for training the data. Only parameters that are fundamental for training will be extracted from the dataset. In addition to the these datasets used for training, a separate test dataset is constructed to test the model on an entirely new domain not present during training.

B. Data Preprocessing

The main goal in preprocessing is to ensure that the data is clean before merging the dataset into balanced rows between fake and real reviews. To obtain this consistency, removing duplicates, special symbols and tokenizing the words is necessary, and done using the XLM-RoBERTa multilingual tokenizer, which converts text into numbers in order to be consistent across different languages. This helps the model understand the semantics of the words in different languages and be able to properly analyze. The class labels of the datasets

do not have the same values. Therefore, the labels of the merged dataset have been cleaned and the values obtained are 1 for a fake review and 0 for a real one. Our model handles multiple languages as well as multiple domains, hence, for each tuple, the dataset should specify in which language the review was written, and the domain the review was written about. The parameters of the newly merged dataset ready for training are:

- 1) Review-Text: which is the actual review content that will be cleaned and tokenized for consistency.
- 2) Label: which is a binary value (1 for Fake, 0 for Real), and it is the target variable for training and evaluation.
- 3) Rating: it is an integer input that is filled either by the actual rating in the review and ranges from 0 to 5, or by a -1 whenever the rating is not specified.
- 4) Language: it is a categorical feature that identifies the language of each review; 'en' for the english language and 'ar' for the arabic language.
- 5) Domain: it is a categorical feature that is used for cross-domain adaptability.

These standardized parameters ensure that training is consistent across datasets that originally differed in format, language, and features.

C. Model Training

The training dataset that was constructed by merging several publicly available review sources, will be randomly split by 80/20, for the training and for the testing. The XLM-RoBERTa is the model used as a base for the CME, and it is a multilingual transformer model pre-trained on over 100 languages. A Softmax classification layer is added on top of the transformer to perform accurate binary classification (fake vs. real). The model is fine-tuned on the multilingual datasets using the AdamW optimizer, with a learning rate of $2e-5$, batch size of 16, and trained for 2 epochs. The binary cross-entropy loss function is minimized during training to improve accuracy and stability. This training process evaluates whether multilingual and cross-domain representations learned by CME can adapt effectively to new types of reviews.

D. Model Testing & Evaluation

After fine-tuning the model, CME is tested on datasets to assess its ability to generalize across languages and domains. To test CME's robustness and to see how well it can adapt and stay accurate, a separate testing dataset was created. This dataset contains english reviews covering a completely new domain, unseen during training, including real and fake reviews from clothing platforms. This approach measures true multilingual and cross-domain generalization, which is an essential requirement for a model designed to operate across diverse online platforms. This enables a clearer understanding of whether CME can maintain accuracy when new domains are introduced, and also provides a true zero-shot evaluation, since the model has never been exposed to these inputs during training. The CME model will then be evaluated on key performance metrics: Accuracy, Precision, Recall, and

TABLE II
SYSTEM PARAMETERS

Parameter	Value
Languages	English, Arabic
Dataset Size	39,913
New Domain Dataset Size	3,849
Data Split	80% Training, 20% Testing
Model Architecture	XLM-RoBERTa-base (Transformer)
Loss Function	Binary Cross-Entropy
Optimizer	AdamW
Learning Rate	$2e-5$
Batch Size	16
Epochs	2
Evaluation Metrics	Accuracy, Precision, Recall, F1-score
Platform	Google Colab (GPU)

F1-score. Comparative analysis will be conducted against baseline models, like COAT [4], to check for improvements in cross-domain adaptability, multilingual understanding, and interpretability.

E. Explainability

Instead of external explainability tools like SHAP, CME uses its built-in attention mechanism from the XLM-RoBERTa to visualize which words had the most influence on the classification decision. The attention scores are extracted from the final layers of XLM-RoBERTa and transformed into bar plot using Matplotlib. This allows users to see which terms, such as "perfect service" or "worst ever," contributed most to labeling a review as fake or real. This layer adds transparency for users and gains trust for the model.

V. RESULTS AND ANALYSIS

A. Simulation Setup

The CME model was implemented using the XLM-RoBERTa-base architecture and trained on a multilingual and multidomain datasets combining English and Arabic fake and real reviews across hotel, restaurant, and e-commerce domains. After preprocessing and balancing, the merged dataset consisted of 39,913 samples, split into 80% for training and 20% for internal testing. Training was conducted on Google Colab using an NVIDIA T4 GPU. The model was fine-tuned for two epochs using a batch size of 16, a learning rate of 2×10^{-5} , the AdamW optimizer, and Binary Cross-Entropy loss. To evaluate domain generalization, an external *Clothing Review Dataset* containing 3,849 reviews was used as a zero-shot test set since it represents a domain not included in training. Explainability analysis was conducted using an attention-based explainability layer. Error analysis was further performed using a confusion matrix on sample predictions. The evaluation metrics follow

standard definitions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

B. Performance Results and Analysis

1) *Test Set (Seen Domains and Languages)*: The 20% test set (7,983 samples) was used to evaluate performance on domains and languages observed during training. The performance of the model is recorded in the Table III. These results show that CME effectively models deceptive-review patterns in multilingual and multidomain settings. A confusion matrix was created using Matplotlib to determine these results.

TABLE III
TEST SET PERFORMANCE (20% SPLIT)

Metric	Accuracy	Precision	Recall	F1
Value	89%	0.84	0.94	0.89

2) *Zero-Shot Cross-Domain Evaluation (Clothing Domain)*: The Clothing Review Dataset (3,848 samples) evaluates the model’s domain generalization capability in a fully unseen category. Despite the domain shift, CME maintained an accuracy of **70%**. The confusion matrix showed 1,320 true positives, 1,368 true negatives, 600 false positives (real reviews wrongly predicted as fake), and 560 false negatives (fake reviews wrongly predicted as real). The TP and TN rates are higher which proves the model was trained efficiently and predicted most labels correctly. The FP/FN rate shows that more real reviews were incorrectly predicted as fake, which is useful for fraud prevention but can penalize genuine reviewers. These findings in Figure 2 confirm that the model transfers effectively to new product categories without retraining.

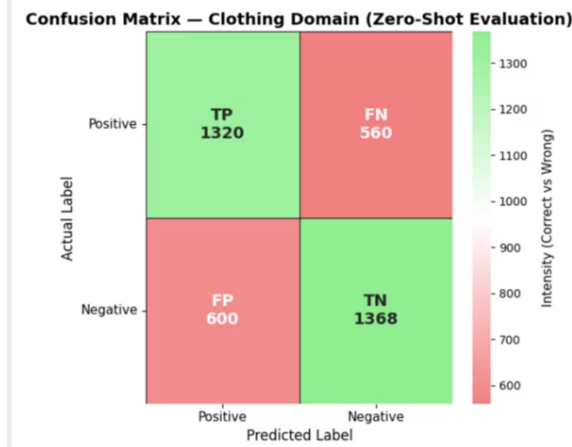


Fig. 2. Confusion Matrix Heatmap for Zero-Shot Clothing Domain Evaluation.

3) *Explainability Analysis*: To analyze model interpretability, a sample review from the dataset chosen at random; “I have had mine for two years now. Very reliable and extremely

useful.” was visualized with an attention-based explainability bar chart in Fig.3, using CLS-to-token attention weights. Words such as “very”, “reliable”, “extremely”, and “useful” displayed the highest contributions, confirming that CME relies on meaningful linguistic cues and semantics rather than superficial lexical patterns.

4) *Comparison with Existing Work*: We compare CME with COAT, a state-of-the-art cross-domain fake review detector evaluated on Amazon domains. Although COAT also performs cross-domain fake review detection, its experiments are limited to English-only datasets. In contrast, our CME model extends cross-domain generalization into a multilingual setting by combining Arabic and English in the training dataset. Additionally, COAT operates as a black-box classifier with no built-in interpretability, while CME integrates attention-based explainability, allowing the model to highlight the specific tokens influencing each prediction. COAT achieves on average 68% accuracy across cross-domain settings, whereas CME achieves **70%** on a fully unseen domain.

TABLE IV
COMPARISON OF COAT AND CME ON CROSS-DOMAIN METRICS

Model	Accuracy	Precision	Recall	F1-score
COAT [4]	67.74%	68.82%	70.24%	69.83%
CME	70%	69%	68%	69%

VI. CONCLUSION

This research presented CME, a cross-domain and multilingual fake-review detector built on XLM-RoBERTa. CME unifies three elements often treated separately: learning from multilingual and multidomain datasets, achieving strong generalization without domain-specific retraining, and providing interpretable predictions through attention-based token attribution. Trained on merged English–Arabic reviews, the model reached 89% accuracy on the internal 20% test set and 70% accuracy in a zero-shot clothing domain, outperforming COAT, which reports 68% on similar cross-domain tasks. Attention visualizations confirmed that CME relies on meaningful semantic cues.

Despite these strengths, CME was evaluated on only two languages and shows a tendency toward higher recall, which may over-flag genuine reviews. Attention maps also offer only partial insight into the model’s internal reasoning. Future work includes expanding language coverage, improving interpretability, and incorporating additional diverse domains.

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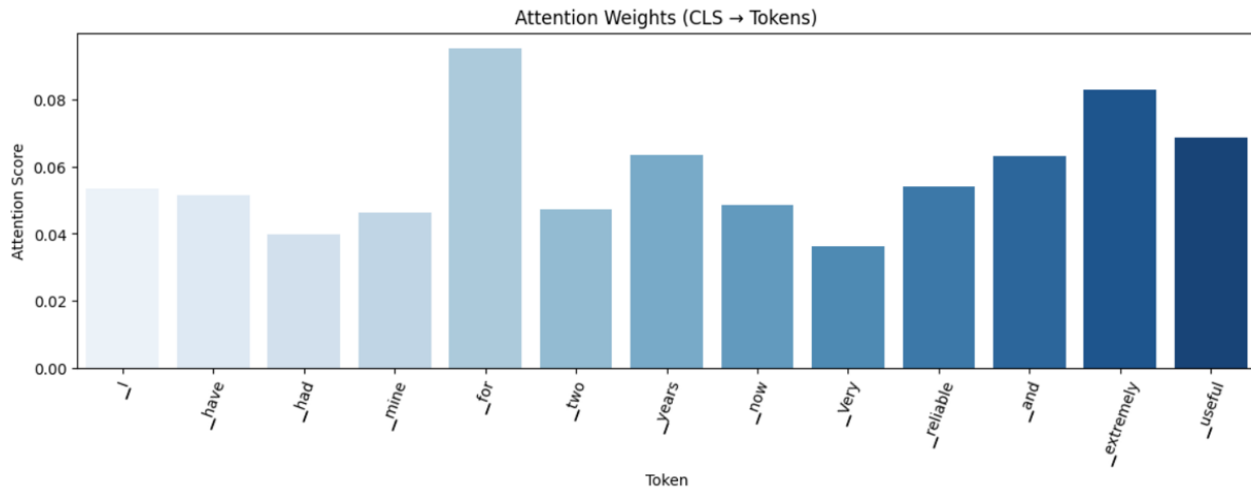


Fig. 3. Attention-Based Token Importance Visualization for Example Review. The visualization highlights the model's CLS-to-token attention distribution, emphasizing the most influential words used during prediction.

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